

Research Notes on F1 score

F1 score

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

>> Section 1

Extension to multi-class classification The F-score is also used for evaluating classification problems with more than two classes (Multiclass classification). A common method is to average the F-score over each class, aiming at a balanced measurement of performance.

>> Section 2

$$F_1 = \frac{2 \cdot \text{TP}}{2 \cdot \text{TP} + 2 \cdot \text{FN}} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad \left\{ \begin{array}{l} \text{displaystyle} \\ F_1 = \frac{2 \cdot \text{TP}}{2 \cdot \text{TP} + 2 \cdot \text{FN}} \\ \text{FN} \end{array} \right\}$$

>> Section 3

In statistical analysis of binary classification and information retrieval systems, the F-score or F-measure is a measure of predictive performance. It is calculated from the precision and recall of the test, where the precision is the number of true positive results divided by the number of all samples predicted to be positive, including those not identified correctly, and the recall is the number of true positive results divided by the number of all samples that should have been identified as positive. Precision is also known as positive predictive value, and recall is also known as sensitivity in diagnostic binary classification. The F1 score is the harmonic mean of the precision and recall. It thus symmetrically represents both precision and recall in one metric. The more generic F_β score applies additional weights, valuing one of precision or recall more than the other. The highest possible value of an F-score is 1.0, indicating perfect precision and recall, and the lowest possible value is 0, if the precision or the recall is zero.

>> Section 4

Two commonly used values for β are 2, which weighs recall higher than precision, and 1/2, which weighs recall lower than precision. The F-measure was derived so that F_β "measures the effectiveness of retrieval with respect to a user who attaches β times as much importance to recall as precision". It is based on Van Rijsbergen's effectiveness measure

>> Section 5

$$F_1 = \frac{2 \cdot \text{TP}}{2 \cdot \text{TP} + 2 \cdot \text{FN}} = \frac{2 \cdot \text{TP}}{2 \cdot \text{TP} + 4 \cdot \text{TP}} = \frac{1}{2} = 0.5$$

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>> Section 6

Macro F1 Macro F1 is a macro-averaged F1 score aiming at a balanced performance measurement. To calculate macro F1, two different averaging-formulas have been used: the F1 score of (arithmetic) class-wise precision and recall means or the arithmetic mean of class-wise F1 scores, where the latter exhibits more desirable properties.

>> Section 7

$$F_1 = \frac{2 \cdot \text{recall} \cdot \text{precision}}{\text{recall} + \text{precision}} = \frac{2 \cdot \text{TP}}{2 \cdot \text{TP} + \text{FP} + \text{FN}}$$

>> Section 8

Micro F1 Micro F1 is the harmonic mean of micro precision and micro recall. In single-label multi-class classification, micro precision equals micro recall, thus micro F1 is equal to both. However, contrary to a common misconception, micro F1 does not generally equal accuracy, because accuracy takes true negatives into account while micro F1 does not.

>> Section 9

$$F_1 = \frac{2 \cdot \text{TP}}{2 \cdot \text{TP} + 2 \cdot \text{FP}} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

>> Section 10

To see it as a harmonic mean, note that $F_1 = \frac{2 \cdot \text{recall} \cdot \text{precision}}{\text{recall} + \text{precision}}$

>> Section 11

F_β score A more general F score, F_β , that uses a positive real factor β , where β is chosen such that recall is considered β times as important as precision, is:

>> Section 12

r of positive to negative test cases. This means that comparison of the F-score across different problems with differing class ratios is problematic. One way to address this issue (see e.g., Siblani et al., 2020) is to

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use a standard class ratio r_0 when making such comparisons.

>> Section 13

$$F_\beta = \beta^2 + 1 (\beta^2 \cdot \text{recall} - 1) + \text{precision} - 1 = (1 + \beta^2) \cdot \text{precision} \cdot \text{recall} (\beta^2 \cdot \text{precision} + \text{recall})$$
$$F_{\beta} = \frac{\beta^2 + 1}{\beta^2 \cdot \text{recall} + \text{precision}}$$
$$F_{\beta} = \frac{(1 + \beta^2) \cdot \text{precision} \cdot \text{recall}}{\beta^2 \cdot \text{precision} + \text{recall}}$$

>> Section 14

With $\text{precision} = \text{TP} / (\text{TP} + \text{FP})$ and $\text{recall} = \text{TP} / (\text{TP} + \text{FN})$, it follows that the numerator of F1 is the sum of their numerators and the denominator of F1 is the sum of their denominators. If $\text{FP} = \text{FN}$

>> Section 15

To see that as a weighted harmonic mean, note that $F_\beta = \frac{1}{\frac{1}{\beta^2 + 1} + \frac{1}{\beta^2 \cdot \text{recall} + \text{precision}}}$. In terms of Type I and type II errors this becomes:

>> Section 16

$$F_1 = \frac{2 \text{TP}}{2 \text{TP} + 2 \text{FP}} = \frac{2 \text{TP}}{4 \text{TP}} = \frac{1}{2} = 0.5$$
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>> Section 17

Applications The F-score is often used in the field of information retrieval for measuring search, document classification, and query classification performance. It is particularly relevant in applications which are primarily concerned with the positive class and where the positive class is rare relative to the negative class. Earlier works focused primarily on the F1 score, but with the proliferation of large scale search engines, performance goals changed to place more emphasis on either precision or recall and so F_β is seen in wide application. The F-score is also used in machine learning. However, the F-measures do not take true negatives into account, hence measures such as the Matthews correlation coefficient, Informedness or Cohen's kappa may be preferred to assess the performance of a binary classifier. The F-score has been widely used in the natural language processing literature, such as in the evaluation of named entity recognition and word segmentation.